Probability

The big idea

Probability measures how likely an event is, on a scale from **0 (impossible)** to **1 (certain)**. In this chapter we use **theoretical (classical) probability**, where all outcomes are **equally likely**.

THEORETICAL PROBABILITY

P(E) = (number of outcomes favourable to E) / (total number of possible outcomes)

FACTS TO REMEMBER

$0 \le P(E) \le 1$. P(sure event) = 1. P(impossible event) = 0.
P(E) + P(not E) = 1, so P(not E) = 1 – P(E). E and "not E" are complementary events.
The sum of the probabilities of all elementary events is 1.

Know your sample spaces

- **Coin**: 2 outcomes (H, T). Two coins: 4 (HH, HT, TH, TT).
- **Die**: 6 outcomes (1–6). Two dice: **36** outcomes.
- **Deck of cards**: 52 cards, 4 suits (spades ♠, clubs ♣ black; hearts ♥, diamonds ♦ red), 13 cards each. Face cards: J, Q, K (12 in all). Aces: 4.

ONE CARD IS DRAWN FROM A WELL-SHUFFLED DECK. FIND P(A FACE CARD) AND P(NOT A HEART)

Face cards = 12, so P = 12/52 = **3/13**.

Hearts = 13, so P(not a heart) = 1 – 13/52 = **3/4**.

TWO DICE ARE THROWN. FIND P(SUM IS 7)

Favourable: (1,6), (2,5), (3,4), (4,3), (5,2), (6,1): 6 outcomes.

P = 6/36 = **1/6**.

A BAG HAS 3 RED AND 5 BLACK BALLS. FIND P(NOT RED)

P(red) = 3/8, so P(not red) = **5/8**.

COUNTING (1,6) AND (6,1) ONCE

With two dice, (1,6) and (6,1) are **different** outcomes: die 1 shows 1 and die 2 shows 6, or the other way round.

Quick recap

- P(E) = favourable / total, for equally likely outcomes.
- 0 ≤ P(E) ≤ 1 and P(E) + P(not E) = 1.
- Know coin, die and card sample spaces by heart.

Practice questions

1. Which of these cannot be a probability?
   (a) 0.7 (b) 2/3 (c) –1.5 (d) 15%
2. If P(E) = 0.05, then P(not E) is
   (a) 0.05 (b) 0.95 (c) 0.5 (d) 1.05
3. A die is thrown once. The probability of getting a prime number is
   (a) 1/2 (b) 1/3 (c) 2/3 (d) 1/6
4. A card is drawn from 52 cards. The probability of an ace is
   (a) 1/52 (b) 1/13 (c) 1/4 (d) 4/13
5. Two coins are tossed. The probability of at least one head is
   (a) 1/4 (b) 1/2 (c) 3/4 (d) 1

Answers: 1. c 2. b 3. a 4. b 5. c